

Problem Set 32

Substitution

1

Read pages 787 - 796 in Gottlieb's Textbook. If you'd like a little extra practice, Exercise 25.2 on page 796 gives you some hints without giving all of the details. You can also try the problems on the in-class worksheet that you didn't get to in class (solutions are posted on the course website).

Evaluate the following integrals using the given substitution. (Can you see why the given substitution is a good choice?)

Check your answers to (a) and (c) by differentiating them.

(a) $\int x \sin(x^2) dx$, substituting $u = x^2$

(b) $\int_0^1 \frac{1}{(4x+1)^3} dx$, substituting $u = 4x+1$

(c) $\int \frac{x}{x^2+9} dx$, substituting $u = x^2+9$

(d) $\int_0^{\ln 4} e^x \sqrt{3e^x+13} dx$, substituting $u = 3e^x+13$

2

In one of the following integrals, the substitution $u = e^{2x} + 1$ works well; in the other, the substitution $u = e^x$ does (it's up to you to decide which is which). Evaluate both integrals, and check your answers by differentiating them.

(a) $\int \frac{2e^x}{e^{2x}+1} dx$

(b) $\int \frac{e^{2x}}{e^{2x}+1} dx$

3

(parts of 25.2.3, 25.2.4) Evaluate the following integrals. (It may take you a few tries to pick a useful u ; that's normal when you're first learning substitution!)

(a) $\int \frac{1}{\sqrt{2x+1}} dx$

(b) $\int \frac{1}{2x+1} dx$

(c) $\int x\sqrt{2x^2+1} dx$

(d) $\int \frac{x}{\sqrt{2x^2+1}} dx$

4

(parts of 25.2.4, 25.2.6, 25.2.18) Evaluate the following integrals. Pay attention to whether you are working on an indefinite integral or a definite integral and whether your final answer makes sense.

$$(a) \int_{1/4}^1 \frac{\cos(\sqrt{x})}{\sqrt{x}} dx \quad (b) \int \sqrt{\cos x} \sin x dx \quad (c) \int \tan x dx \quad (d) \int_0^{\ln 5} \frac{3e^x}{\sqrt{e^x + 4}} dx$$

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Reflect Back.

In which of the following integrals could you make a substitution to turn the integral into a constant multiple of $\int \sqrt{u} du$? What substitution would that be?

Example: Substituting $u = \sin x$ in $\int 2\sqrt{\sin x} \cdot \cos x dx$ gives $\int 2\sqrt{u} du$.

$$(a) \int \sqrt{\tan x} dx \quad (c) \int e^x \sqrt{e^{2x} + 1} dx \quad (e) \int \sqrt{3x + 1} dx$$

$$(b) \int \sqrt{\tan x} \cdot \sec^2 x dx \quad (d) \int e^{2x} \sqrt{e^{2x} + 1} dx$$

Hint: You can make an appropriate substitution in three of the five integrals.